CSC 163: Computing, Ethics, and Society

ABET Student Outcome 4 (SO4)

Student guide and instructor workflow

Roadmap

Part A: For students

1. What SO4 is and why we assess it
2. Four performance indicators (4a–4d) and where they are assessed
3. What strong answers contain and how they are scored

Part B: For the instructor (end-of-semester ABET work)

4. Timeline: now, December, January
5. Assessment items 4a–4d and four-level scoring descriptors
6. Filling the rubric sheets, keeping evidence, closing the loop

Part A: For Students

What SO4 asks of you and how your work is scored

ABET and Student Outcome 4

ABET accredits our computing program. To keep accreditation, the program must show evidence that graduates attain a set of **student outcomes**.

Student Outcome 4 (SO4): Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

In CSC 163, three final-exam questions and one report are also scored on a common 1–4 rubric. ABET looks at program-level results, which are used to improve the course.

Four Performance Indicators (PIs)

PI	You should be able to…	Where it is assessed
4a	Describe situations where using software is illegal, and who is harmed	Final exam, short answer (10 pts)
4b	Apply copyright and patent criteria to a software dispute	Final exam, scenario question (10 pts)
4c	Explain, compare and recommend data privacy methods	Privacy Report, 2–3 pages (due Week 12–13)
4d	Identify positive and negative effects of a computing solution on society	Final exam, short answer (10 pts)

What Strong Answers Contain

4a Illegal use: two clearly distinct situations; the legal basis (copyright law or license terms); who is harmed.

4b IP criteria: separate criteria for copyright (expression, substantial similarity, access) and patents (claims, prior art); a step-by-step decision process; defenses such as fair use or clean-room design.

4c Privacy methods: for each of two methods, how it works, which threats it stops and its limits; then compare them and recommend one for a specific application.

4d Societal effects: at least one positive and one negative effect; who is affected in each case; several stakeholder perspectives and trade-offs.

The Four-Level Scale

Level	Label	What it means
1	Unsatisfactory	Fails to demonstrate the skill: missing, off-topic or incorrect
2	Marginal	Limited or less than adequate: partial, vague or one-sided
3	Satisfactory	Appropriate and accurate; covers every required part
4	Excellent	Appropriate and accurate, with depth: precise reasoning, nuance, several perspectives

Aim for **level 3 or above** on every indicator.

Part B: For the Instructor

Planning with the end in mind: decide early, collect and analyze at the end

Timeline

Now (October)

- Finalize questions 4a–4d and the rubric descriptors
- Tell the coordinator: 4a, 4b, 4d = Final Exam Q#; 4c = Privacy Report

December

- Grade the exam and report
- Draw a random sample
- Fill the four sheets; compute the two percentages
- Save sample answers

January

- Submit sheets and samples to the coordinator
- Add a short reflection

Privacy Report planned due date: Week 12–13.

Programs often assess each outcome on a multi-year cycle (for example, once every three years), so this may not recur every semester. Confirm the cycle with the coordinator.

Assessment Items: 4a and 4b

4a Final Exam, short answer (10 pts)

Describe two distinct situations in which the use of software is illegal (for example, under copyright law or a software license agreement). For each situation, (i) explain what makes the use illegal, and (ii) identify who is harmed.

4b Final Exam, scenario question (10 pts)

A start-up releases a mobile app whose user interface, source code structure, and a key algorithm closely resemble those of an existing commercial product. Describe the criteria and steps you would use to decide whether the start-up has violated the other company's copyright and/or patent. Be explicit about which criteria apply to copyright and which apply to patents.

Assessment Items: 4c and 4d

4c Privacy Report (2–3 pages, due Week 12–13)

Select two methods used to safeguard data privacy (for example, encryption at rest or in transit, anonymization or pseudonymization, access control, data minimization, differential privacy). For each method: describe how it works, what threats it protects against, and its limitations. Then compare the two methods and recommend which is more appropriate for a specific application you choose, justifying your recommendation.

4d Final Exam, short answer (10 pts)

Choose one computing solution (for example, facial recognition in public spaces, ride-sharing platforms, automated résumé screening, generative AI tools). Identify at least one positive and one negative effect of this solution on society, and explain who is affected in each case.

Scoring Descriptors: 4a and 4b

PI	1 Unsatisfactory	2 Marginal	3 Satisfactory	4 Excellent
4a Illegal software use	Cannot describe a valid illegal-use situation, or the examples are not illegal	Describes one valid situation, or two with vague or incorrect reasons	Two valid situations with the correct legal basis; identifies the harmed party	Two clearly distinct situations (e.g., piracy vs. license violation), precise legal basis, nuanced discussion of harm
4b IP violation criteria	No usable criteria; confuses copyright and patents entirely	Lists some criteria but mixes copyright and patent, or omits a decision process	Correct criteria for copyright (expression, substantial similarity, access) and patents (claims, prior art) and a reasonable decision process	Correct, well-structured criteria for both; applies them to the scenario; notes fair use, clean-room or independent invention

Scoring Descriptors: 4c and 4d

PI	1 Unsatisfactory	2 Marginal	3 Satisfactory	4 Excellent
4c Data privacy methods	Fewer than two methods, or methods are misdescribed	Names two methods but the description is superficial; no real analysis or comparison	Correctly explains two methods, the threats addressed and the limitations; makes a comparison	Deep analysis of mechanism, threat model and limits; well-justified, context-specific recommendation
4d Societal effect	No identifiable effect, or the effect is unrelated to the solution	Identifies an effect but it is one-sided, or does not explain who is affected	At least one positive and one negative effect, and the affected groups	Several well-articulated effects, several stakeholder perspectives, considers trade-offs

Filling the Rubric Sheets (December)

1. **Where Assessed** and date: e.g., “Final Exam, Question 5” or “Privacy Report, Part 2”
2. Draw a random sample of students (anonymized IDs are fine); follow the sample size the coordinator specifies
3. Score each sampled student 1–4 on each PI using the descriptor table
4. Compute the **average** score
5. Compute two percentages: **% scoring 1** (target ≤ 10%) and **% scoring 3 or 4** (target ≥ 70%)

Illustrative example (10 sampled scores): 4, 3, 3, 2, 3, 4, 1, 3, 2, 3
Average = 28 / 10 = 2.8; % at level 1 = 1/10 = 10% (met); % at level 3–4 = 7/10 = 70% (met)

Evidence to Keep

Keep these for the coordinator and for the eventual ABET visit:

- A PDF of each assessment question (4a–4d)
- The four-level descriptor table (slides 12–13)
- The sampled student answers, ideally one at each of levels 1–4 for each PI
- The completed rubric sheets with averages and both percentages

Closing the Loop

If a PI misses its target, add **two or three sentences** on what you will change in the next offering. Closing the loop is what ABET evaluators look for.

- **Finding:** what the data showed (which PI, how far from target)
- **Likely cause:** what in the course or the question explains it
- **Action:** what changes next offering

Illustrative: 4b had 55% at level 3–4 (target 70%). Many answers mixed copyright and patent criteria. Next offering: add a worked in-class example and a practice question that separates the two.